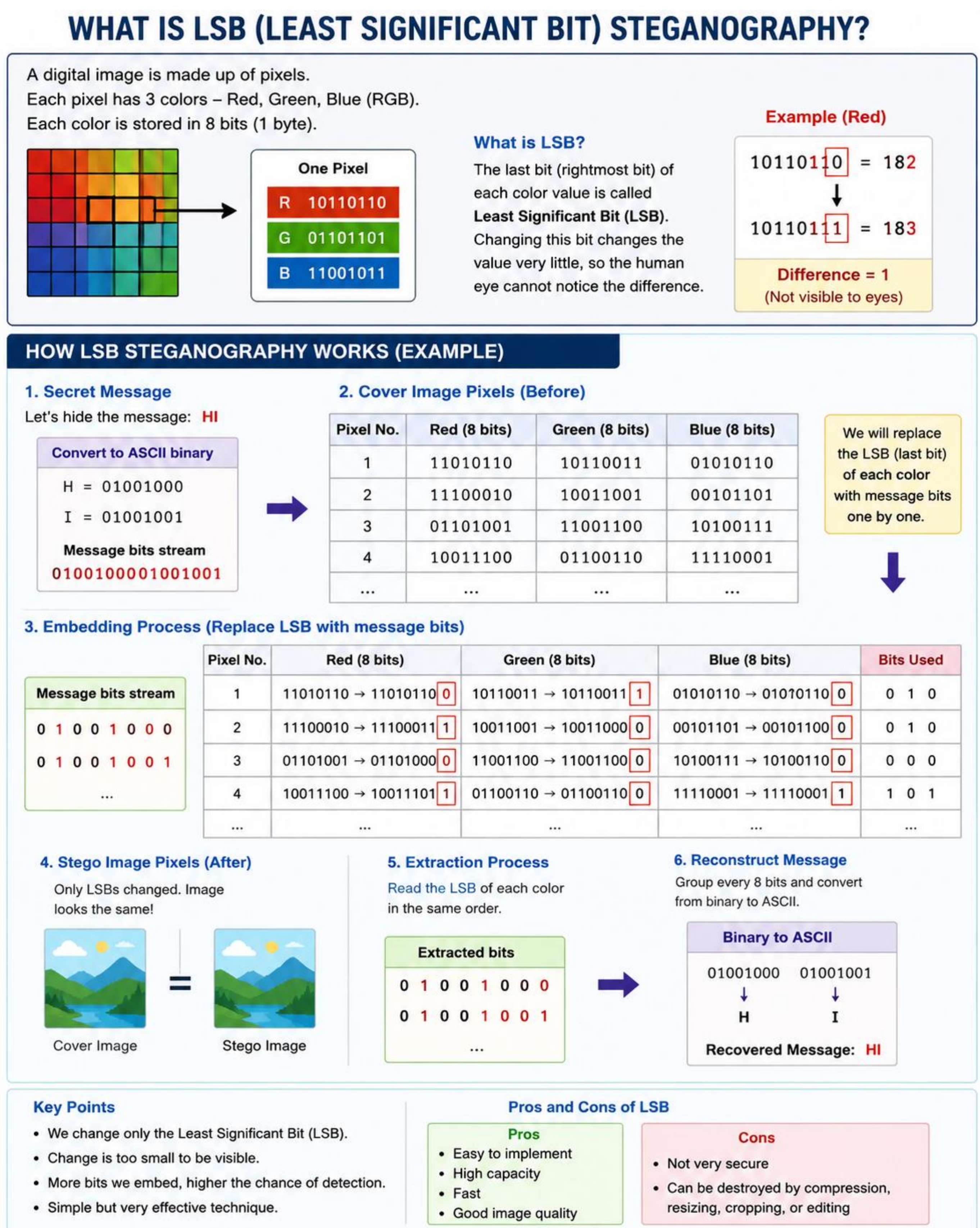
Img-Crypt v2 Documentation

Contents

1: Introduction

Img-Crypt is an open-source command-line steganography utility developed in Python. It enables users to hide (encode) and retrieve (decode) secret text messages inside image files using Least Significant Bit (LSB) steganography. The tool is designed with simplicity in mind, making it suitable for students, cybersecurity learners, CTF participants, & digital forensics enthusiasts.

Unlike many traditional steganography tools that rely entirely on command-line arguments, Img-Crypt provides an interactive menu-driven interface while still supporting encryption of hidden messages.



2: Objectives

The objectives of Img-Crypt are:

- Hide secret messages inside images.
- Retrieve hidden messages from images.
- Protect hidden messages using password-based encryption.
- Provide a beginner-friendly interface.
- Support Windows and Kali Linux.
- Serve educational and research purposes.

3: Features

Message Encoding

- Hide text inside PNG or BMP images.
- Manual text input.
- Load text from a TXT file.
- Image size validation.
- Output image generation.

Message Decoding

- Extract hidden text.
- Detect encrypted messages automatically.
- Password verification.
- Save long messages to a text file.

Security

- Optional password protection.
- Fernet symmetric encryption.
- Base64 key generation.

User Experience

- Colored terminal interface.
- Animated disclaimer.
- KeyboardInterrupt handling.
- Error handling.
- Cross-platform compatibility.

4: Advantages

- Easy to use.
- Interactive.
- Password protection.
- Open source.
- Lightweight.
- Cross-platform.
- Educational.
- Beginner friendly.

5: Limitations

- Only text messages.
- No audio steganography.
- No video steganography.
- No PDF/document hiding.
- Single-image support.
- Uses only LSB.
- No metadata hiding.
- No compression.
- No steganalysis detection.